

Module Design

1. Main Module

File: `main.py`

Responsible for creating the FastAPI application and loading the application routes and templates.

2. Routes Module

File: `routes.py`

Handles the main application routes:

- `/`
- `/generate-workout`
- `/submit-feedback`
- `/view-all-users`

The project source defines these four core routes.

3. Workout Generator

File: `geminigenerator.py`

Function:

`generate_workout_gemini()`

Generates a personalized 7-day workout plan.

4. Nutrition Generator

File: `geminiflashgenerator.py`

Function:

`generate_nutrition_tip_with_flash()`

Generates a practical nutrition or recovery tip.

5. Updated Plan Module

File: `updated_plan.py`

Function:

`update_workout_plan()`

Uses the original plan and user feedback to generate an updated plan.

6. Database Module

File: `database.py`

Stores user information and workout plans using SQLite and SQLAlchemy.

7. Schema Module

File: `schemas.py`

Contains the data structures used for validating user input and feedback.